

Same synergy, different added information

Exact analytic example · signed net MGW atoms in nats

A, B, U independent fair bits; target $Y = (A, B)$; baseline $X_1 = A$.

$$\ell = \ln 2 \approx 0.693147; k = \ln(4/3) \approx 0.287682$$

Candidate	Atom	Signed magnitude	Exact	$\text{CMI} = I(X_2; Y \mid X_1)$
Unrelated $X_2 = U$	R		k	CMI: 0 Expected gain: 0
	U_1		$\ell - k$	
	U_2		$-k$	
	S		k	
Completing $X_2 = B$	R		k	CMI: ℓ Expected gain: $0.9\ell \approx 0.623832$
	U_1		$\ell - k$	
	U_2		$\ell - k$	
	S		k	
Duplicate $X_2 = A$	R		ℓ	CMI: 0 Expected gain: $0.18\ell \approx 0.124766$
	U_1		0	
	U_2		0	
	S		0	

U and B have the same synergy k , but their added information is 0 versus $\ln 2$.

Gain: $a = 0.8$, $b = 0.9$; observed availability masks independent of data and each other.

Bayes log-loss information gain; learned-policy benefit is unproved. Zero atoms use dots.